03-18 第2篇

阅读理解（说明文） Why Lightning Comes First and Thunder Later

① On a rainy summer evening, many students have the same experience. A **sudden white flash** cuts across the dark sky, and you open your mouth to count — "one, two, three…" — before a heavy sound rolls over the roof. Lightning seems to always come first, and thunder always comes later. Why is that? Are they two different things, or the same thing wearing two faces?

② Scientists tell us that **lightning and thunder are actually born together**. Inside a big storm cloud, tiny pieces of ice crash into each other. These crashes rub the ice and produce **electricity**, just like rubbing a balloon on your hair. Slowly, the top of the cloud gets a "+" charge and the bottom gets a "−" charge. When the difference becomes huge, the cloud has to "fire out" this energy, either to another cloud or to the ground below. That sudden fire is lightning. At the same moment, the air around the lightning is heated to **about 27,000 ℃** — five times hotter than the surface of the sun. The air quickly grows bigger, then quickly grows smaller again, and this big shake of air is **thunder**.

③ So why does our ear hear thunder later than our eye sees the flash? The answer is simple: **light travels much faster than sound**. Light moves at about 300,000 kilometres per second, which means the flash reaches our eyes almost at the same moment it happens. Sound, however, moves only about 340 metres per second. If the storm cloud is 1.7 km above you, the light arrives in far less than a blink, but the thunder needs about 5 seconds to reach your ear.

④ This idea gives us a safe and easy trick. When you see the flash, start counting: "one-thousand, two-thousand, three-thousand…" Each "one-thousand" is about one second. Now divide the number by **3**. That gives you roughly how many kilometres away the lightning is. If you can count 15, the storm is about 5 km away. If the gap between flash and thunder is very short — less than 3 seconds — the storm is close, and you should **go inside right away**.

⑤ Many old stories say that thunder is a big drum in the sky. Science tells a different story: it is just hot air moving fast. But knowing the science does not take the beauty away. Next time a thunder storm passes, try this: stand by a safe window, watch the sky, count the seconds, and listen to the air itself telling you how near the storm has come.

根据文章内容，选择最佳答案。

1. What is the best question to describe the main topic of paragraph ②?

A. Why are storms more common in summer?
B. How are lightning and thunder formed together?
C. How do clouds travel across the sky?
D. Why does rain always follow thunder?

2. What makes the electricity inside a storm cloud?

A. The rain drops hitting the ground.
B. The wind blowing across the cloud.
C. Small pieces of ice crashing and rubbing together.
D. The heat from the sun shining on the cloud.

3. According to the article, why do we see lightning before we hear thunder?

A. Thunder is produced several seconds after the lightning.
B. Lightning is a different natural event from thunder.
C. Light travels much faster than sound.
D. The cloud sends lightning first and thunder later on purpose.

4. If a student counts 9 seconds between a flash and the following thunder, the storm is about _____ away.

A. 0.5 km
B. 3 km
C. 9 km
D. 27 km

5. What is the main purpose of the last paragraph?

A. To tell readers never to look at lightning.
B. To compare old stories with new research and encourage safe observation.
C. To prove that ancient people knew thunder very well.
D. To ask students to stop listening to old sky stories.

答案与解析

1.
【答案】B
【题型】段落大意题
【定位】第②段 "lightning and thunder are actually born together" 及之后的解释
【解析】本段讲闪电与雷一起产生的原因。A、C、D 均不是本段主题。故选 B。

2.
【答案】C
【题型】细节理解题

【定位】第②段 "tiny pieces of ice crash into each other... produce electricity"
【解析】小冰晶相互撞击摩擦产生电。A、B、D 原文未提及。故选 C。

3.
【答案】C
【题型】细节理解题
【定位】第③段 "light travels much faster than sound"
【解析】光速远快于声速。A 与文中同时产生矛盾；B、D 与文章不符。故选 C。

4.
【答案】B
【题型】数值推断题
【定位】第④段 "divide the number by 3"
【解析】9 ÷ 3 = 3，故雷暴约 3 km。A、C、D 计算不符。故选 B。

5.
【答案】B
【题型】写作意图题
【定位】第⑤段对比"天上大鼓"的老故事与科学解释，并鼓励安全观察
【解析】既尊重老故事，又鼓励科学观察。A 表达过度；C、D 与原文不符。故选 B。

词汇注释（超纲词）

- **lightning** [n.] 闪电
- **thunder** [n.] 雷；打雷
- **charge** [n.] 电荷